Comprehensive Bill of Materials (BOM): L-M3D "Abyss" Zero-Risk System

Classification: Universal Transfer Method (MUT) / Industrial Deterministic Standard
Currency: USD (Estimated Prototype Costs)

Component	Technical Specifications & Original Dimensions	Material / Chemistry	Qty	Est. Unit Cost ($)	Total Cost ($)
High-Performance Baseplate	Core Area: 20 x 20 mm ($A = 4 \times 10^{-4}$ m²). Thickness: 0.5 mm. Features: Micro-milled Skived-Fins. Surface coating: 10 µm amorphous plating, hardened to 1000 HV.	CuW 85/15 (Copper-Tungsten) with Ni-P (Nickel-Phosphorus) coating	1	85.00	85.00
Zero-Void Interconnect Paste	Particle size: 20-50 nm. Target layer thickness (d_{Ag}): 20 µm. Pressure-sintered at 10 MPa and 250°C. Yields 0% voids.	Pure Nano-Silver (Ag-Sintering Paste), 10g syringe	1	45.00	45.00
Active Metal Brazed Substrate	Carrier layer for structural integrity and thermal matching. Hermetically sealed base.	Si_3N_4 (Silicon Nitride) Ceramic	1	25.00	25.00
Impingement Plenum Housing	CNC-milled block housing 25 micro-nozzles (Nozzle Ø: 0.5 mm). Integrated 70° Knife-Edges for CF sealing.	AlMg3 Aluminum Alloy	1	180.00	180.00
ConFlat (CF) UHV Gaskets	Ring diameter (D): 30 mm. Penetration depth (b_d): 0.2 mm. Yield strength (σ_Y) ≈ 70 MPa.	OFHC (Oxygen-Free High Thermal Conductivity) Copper	2	7.50	15.00
Magnetic Coupling Pump	Brushless, leak-free design. Flow rate (Q): 2 l/min (3.33×10^{-5}	Industrial Standard Components	1	160.00	160.00

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	m^3/s). Required pressure: > 0.8 bar (0.08 MPa).				
Diaphragm Accumulator	Welded expansion bellows. Internal capacity: 35 ml to compensate for 28 ml oil expansion. Fatigue limit $\sigma_{max} < 500$ MPa.	Titanium Grade 5 (Ti-6Al-4V)	1	130.00	130.00
Dielectric Coolant	Dielectric strength: > 40 kV/mm. Volume: 2 Liters ($V_0 = 1000$ ml per system fill). Density (ρ_{oil}): ≈ 850 kg/m^3.	Paraffinum Liquidum (High-purity Mineral Oil)	1	35.00	35.00
Hydrocyclone Separator	CNC-milled bypass module. Separation efficiency: $d_{50} \leq 1$ µm (via Stokes' law centrifugal force).	AlMg3 / Industrial Standard	1	40.00	40.00
Moisture Getter	Inline bypass filtration capsule. Pore size: 4 Å (Angstrom) to trap water molecules.	Zeolite A (Molecular Sieve)	1	15.00	15.00
Passive Diffuser Radiator	Total effective surface area: 2.31 m^2. Height (h): 0.6 m. Features Bell-Mouth Coanda inlet ($\zeta_{in} \approx 0.03$) and 7° flared diffuser ($\zeta_{out} \approx 0.2$).	Extruded Aluminum	1	150.00	150.00
Cylinder Head Screws	Dimensions: M3 x 16 mm. Standard: DIN 912, ISO 4762. Tightening torque (M_A): 0.46 Nm.	Steel 8.8, zinc-plated	4	0.50	2.00
Belleville Washers	Dimensions: 3.2 x 6.0 x 0.4 mm. Standard: DIN 2093. Spring constant (C_f): 4000 N/mm.	Spring Steel 1.4310	4	0.75	3.00
Mounting Backing-Plate	Dimensions: 24 x 24 x 2.0 mm. CNC-milled.	Stainless Steel 1.4301	1	12.00	12.00

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	Distributes 900 N preload force per screw.				
Isolation Washers	Dimensions: M3 flat washer. Ensures galvanic isolation and friction coefficient (μ_k = 0.10).	High-temp PEEK Polymer	4	0.50	2.00
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TOTAL ESTIMATED COST	**Fully assembled Zero-Risk Prototype Unit**	---	---	---	**$ 899.00**